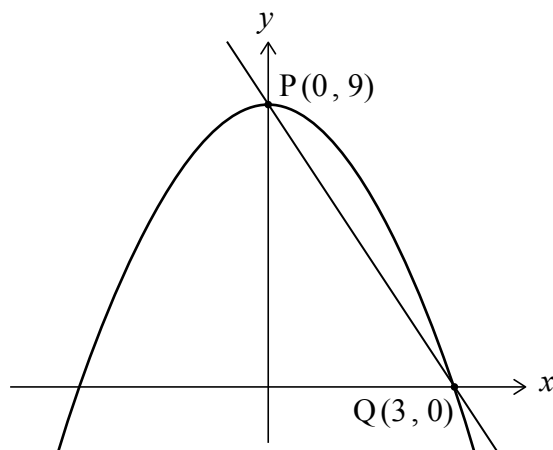


3. [Maximum mark: 7]

A line with equation  $y = -3x + 9$  intersects the axes at the points  $P(0, 9)$  and  $Q(3, 0)$ .

A parabola of the form  $y = ax^2 + c$ , where  $a, c \in \mathbb{Z}$ , also passes through the points  $P$  and  $Q$ .

This is shown in the following diagram.

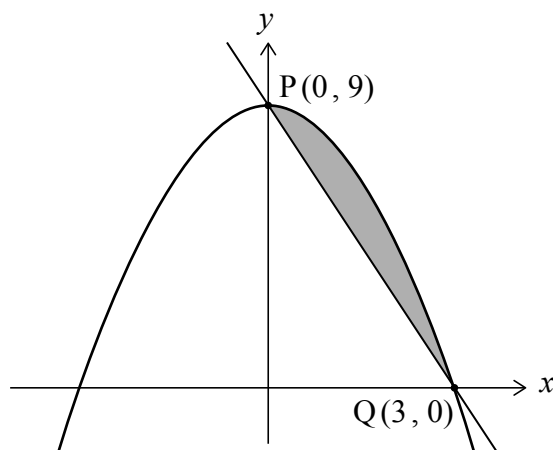


(a) (i) Write down the value of  $c$ .

(ii) Find the value of  $a$ .

[3]

The region enclosed by the line and the parabola is shaded in the following diagram.



(b) Find the area of the shaded region.

[4]

(This question continues on the following page)



(Question 3 continued)

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Turn over